

Lesson 38: Stationary Points & Curve Sketching — Practice Worksheet

Name: _____ Date: _____

Attempt every question. Show your working. The sections increase in difficulty.

Section 1 — Warm-up (foundation)

1. Differentiate $y = x^2 - 6x$.
2. Solve $2x - 6 = 0$.
3. Gradient at a turning point?
4. $f''(x) = 4$: maximum or minimum?
5. $f''(x) = -3$: maximum or minimum?
6. $y = x^2 - 2x$. Stationary x ?

Section 2 — Core practice

7. Find the stationary point of $y = x^2 - 8x + 1$.
8. $y = x^2 + 4x + 1$. Find the stationary point.
9. $y = x^2 - 6x$. Find x at the stationary point.
10. Solve $3x^2 - 12 = 0$.
11. $y = x^2 + 2x$. Find x at the stationary point.

12. $f''(x) = 2$: classify the stationary point.

Section 3 — Challenge & exam-style

13. $y = x^3 - 3x$. Find the stationary points and classify them.

14. $y = x^3 - 12x$. Find the x-coordinates of the stationary points.

15. $y = x^2 - 6x + 5$. Find the stationary point and classify it.

16. $y = 2x^3 - 3x^2$. Find the stationary points.